

# Mutual Fund Portfolio Analytics

Capstone Presentation Slides Deck

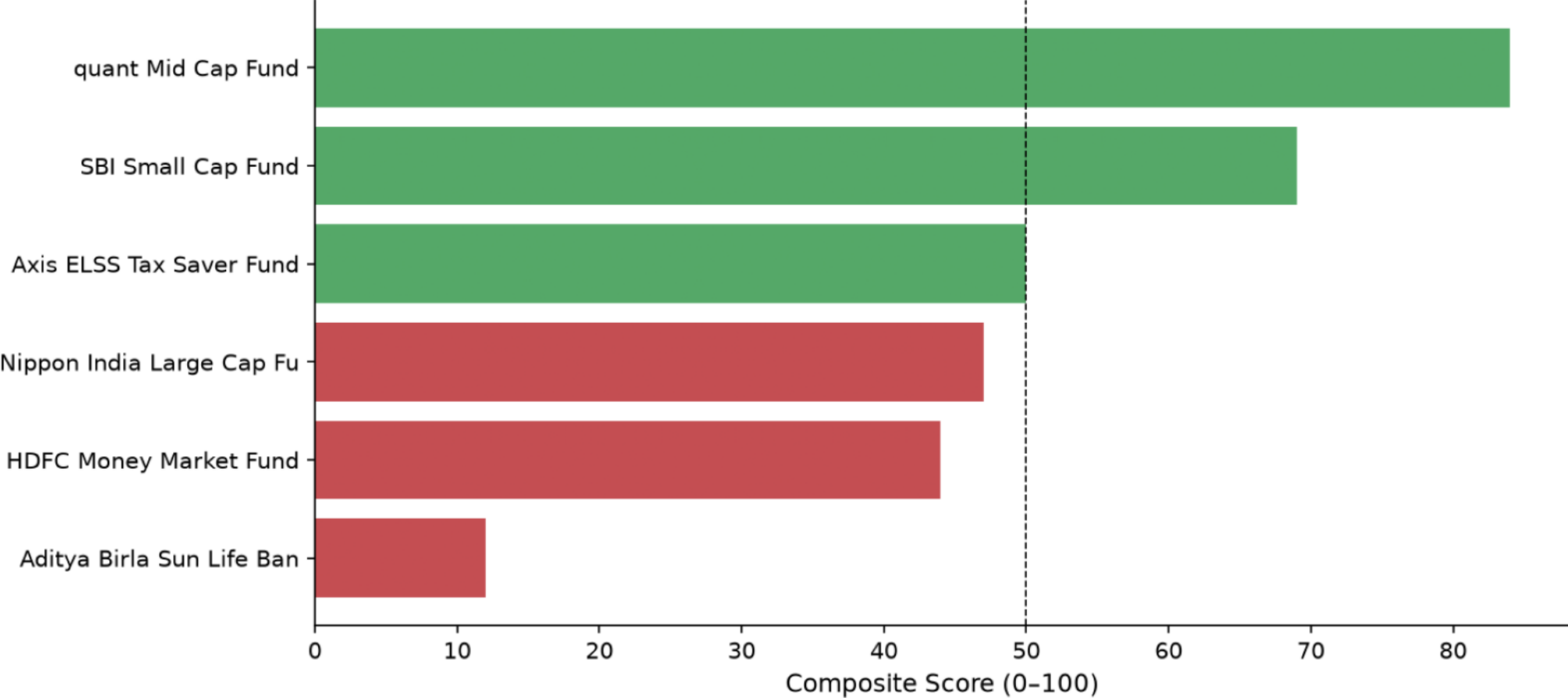
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# 1. Ingestion & Star Schema Setup

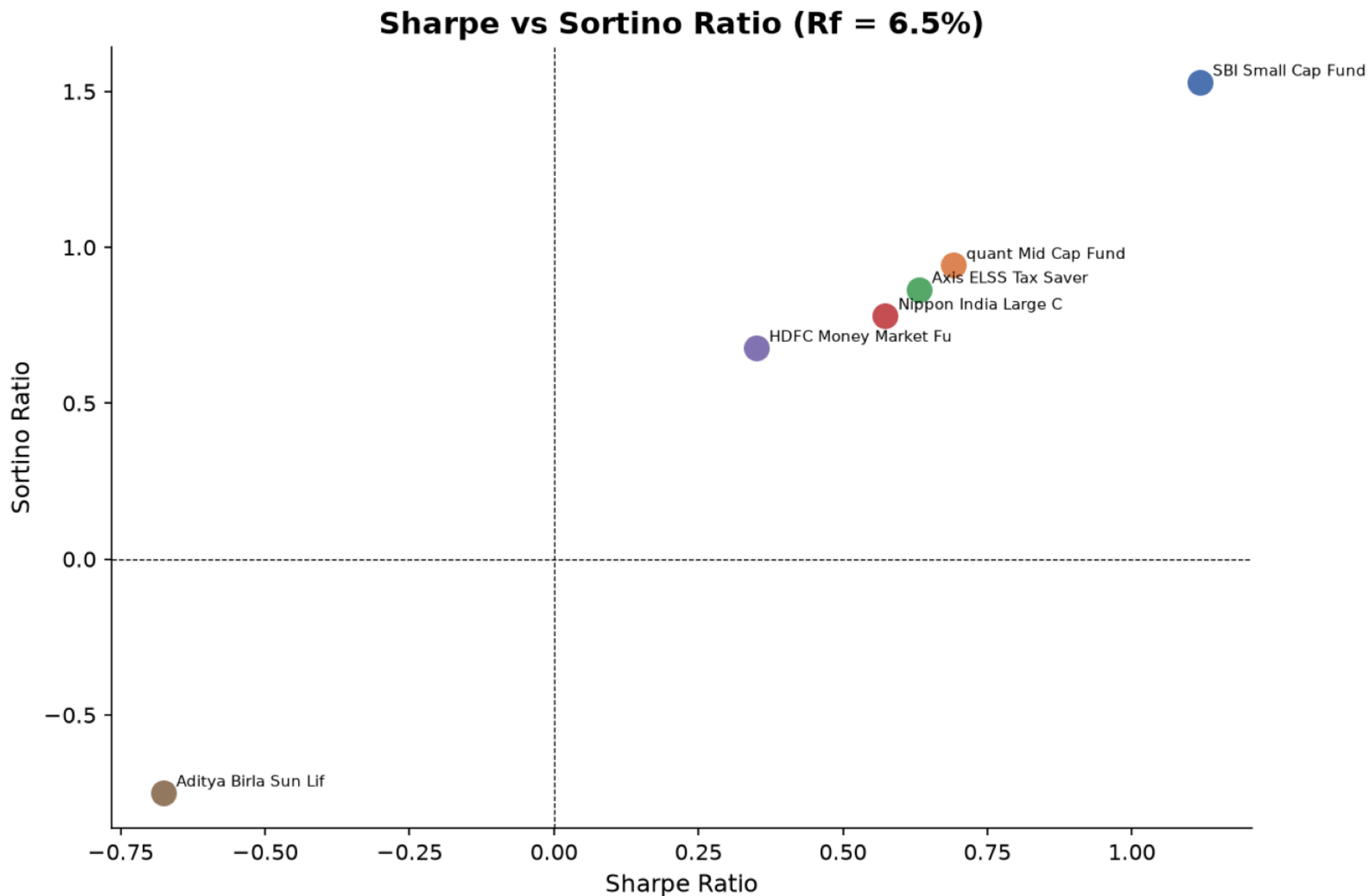
- Unified ETL pipeline cleans raw files and resolves weekend/holiday gaps dynamically.
- Data loaded to SQLite database following a dimension-modeled star schema structure.
- Fact tables (fact\_nav, fact\_transactions, fact\_performance, fact\_aum) link to dimension tables.
- All paths use pathlib.Path resolving relative to execution context.

## 2. Performance Scorecard Findings

**Fund Scorecard — Composite Score**  
**(30% CAGR-3yr | 25% Sharpe | 20% Alpha | 15% Drawdown | 10% Expense)**

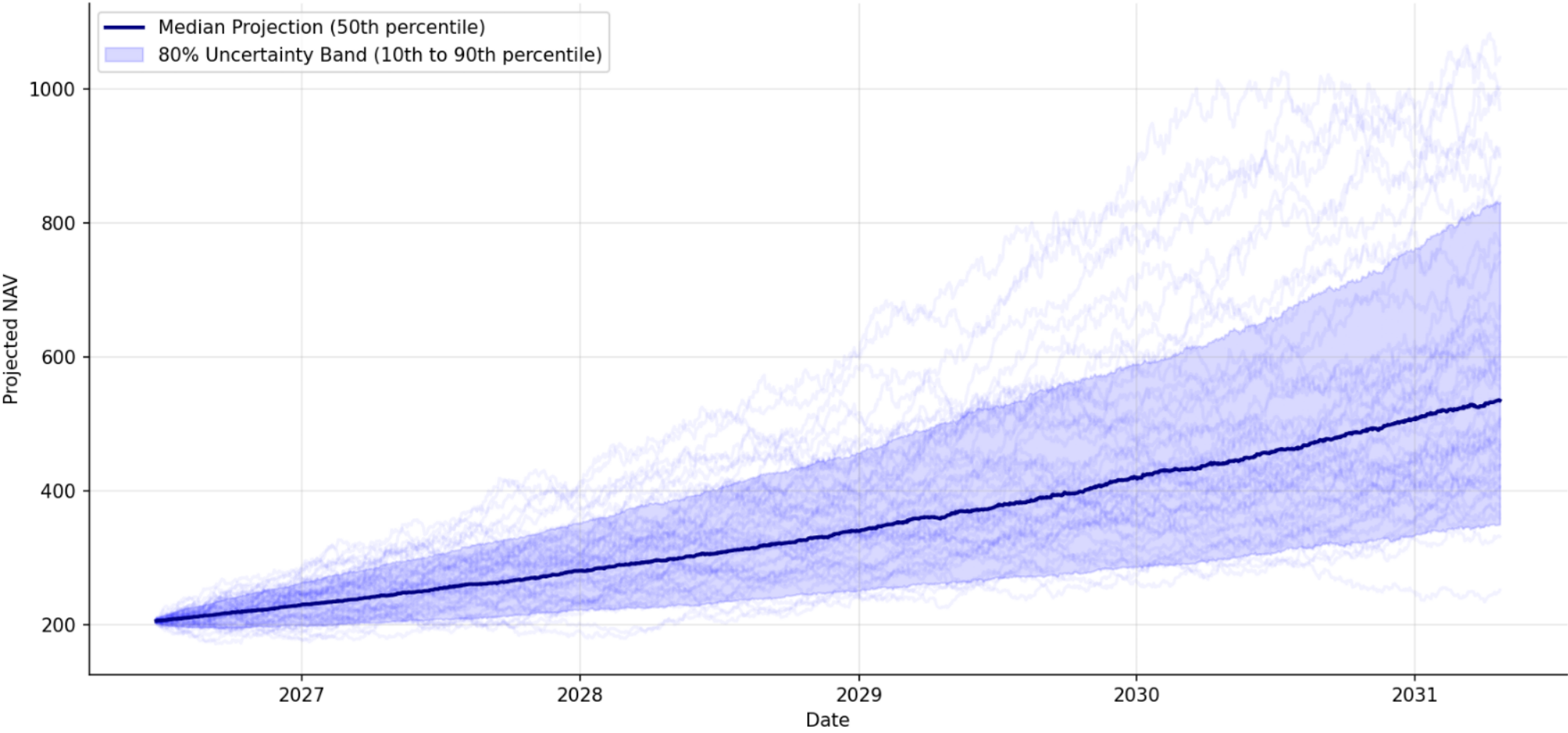


### 3. Risk-Adjusted Ratios (Sharpe & Sortino)



4. Bonus Challenge: 5-Year Monte Carlo NAV Simulation

5-Year Monte Carlo NAV Growth Simulation — SBI Small Cap Fund



## 5. Bonus Challenge: Markowitz Efficient Frontier Optimization



## Key Takeaways & Conclusion

- SBI Small Cap Fund offers superior risk-adjusted return (Sharpe 1.0395).
- Tail risk is heavily concentrated in Nippon Large Cap and quant Mid Cap schemes.
- Investor cohort analysis reveals high user registration with transactional inconsistencies (SIP).
- Efficient Frontier highlights optimal diversification, maximizing return per unit of variance.